



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2017 – 0207



Agricultural Machinery – Rice Combine Harvester FIELDKING FKMCCRHS-2.0

Test Report Number : 2017 – 0207
Valid Until : December 11, 2022
Classification : Commercial
Agricultural Machine : Rice Combine Harvester
Type : Ride-on, crawler-type, sack type, self-propelled
Brand : FIELDKING
Model : FKMCCRHS-2.0
Machine Serial Number : ZZLGE216193
Manufacturer : BERI UDYOG PVT., LTD.
Plot no. 100-101, Sector 3, HSHDC,
Karnal (Haryana), India
Test Requested by : FILHOLLAND CORPORATION
LCG Building, Osmeña St., Cagayan De Oro City
Previous Valid Test Reports : None
of similar Brand and Model

SUMMARY

The FIELDKING FKMCCRHS-2.0 Rice Combine Harvester was tested using “RC 222” rice variety with an average Grain Moisture Content of 24.3% (wet basis) , and average Plant Height of 1175 mm.

Based on the results of the test, the FIELDKING FKMCCRHS-2.0 Rice Combine Harvester had an Actual Field Capacity of 0.619 ha/h or 1.62 h/ha and Field Efficiency of 65.2% at an average Travelling Speed of 4.75 kph. It had a Threshed Grain Output Capacity of 3006 kg/h with threshed grain Purity of 93.2 %. The rice combine harvester had a Total Grain Loss of 0.16 % consisting of 0.06 % Header Loss, 0.10 % Separation/Blower Loss, and negligible Uncut Losses and Unthreshed Losses.

Grain quality analysis of threshed grains showed a net percent Cracked Grain of 1.33 % and Broken Grains of 0.06 %.

The Fuel Consumption Rate of the rice combine’s diesel engine was 7.32 L/h.

The Noise level measured at 50 mm away from the operator’s ear level was 103.1 dB(A).

During the test, three persons were needed to operate the rice combine: one for operating the machine and two for bagging the threshed grain output.

No breakdown or failure of components occurred during the test.

Table 1 shows the performance of the FIELDKING FKMCCRHS-2.0 Rice Combine Harvester in comparison with the Performance Criteria for Rice Combine Harvester in PAES 224:2015.

Table 1. Performance criteria for Rice Combine Harvester

CRITERIA	Performance Requirement from PAES 224:2015	AMTEC Data
Field efficiency, %, minimum	75	65.2
Header loss, %, maximum	0.5	0.06
Separation/Blower loss, %, maximum	2.5	0.10
Unthreshed loss, %, maximum	0.2	0
Scattering loss, %, maximum	0.3	0
Purity, %, minimum	97	93.2
Broken grains, %, maximum	2.0	0.06
Net cracked grain, %, maximum	5.0	1.33
Noise Level, db(A), maximum	95*	103.1

*Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Standards, Department of Labor and Employment, Philippines, 2013.

PURPOSE AND SCOPE OF TEST

The machine was tested for verification of its performance in Brgy. Maligaya, Muñoz, Nueva Ecija and was evaluated on the following items:

1. Field Capacity
2. Field Efficiency
3. Output Capacity
4. Grain losses
 - a. Header Loss
 - b. Separation/Blower Loss
 - c. Uncut Loss
 - d. Unshelled Loss
5. Fuel Consumption

METHOD OF TEST

The FIELDKING FKMCCRHS-2.0 Rice Combine Harvester was tested based on “PNS/PAES 225:2015 – Agricultural Machinery – Rice Combine Harvester – Methods of Test”.

OPERATOR'S MANUAL

No Operator's manual was submitted. Instead, a brochure containing specifications of FIELDKING FKMCCRHS-2.0 Rice Combine Harvester was presented.

DESCRIPTION OF THE MACHINE

The FIELDKING FKMCCRHS-2.0 Rice Combine Harvester was a ride-on, crawler-type, self-propelled harvesting machine which performs both cutting and threshing operations. It was powered by a 65.6 kW diesel engine which drove the transmission systems of the machine. The rice combine harvester consisted of a reel assembly, cutting assembly, screw and chain conveyors, threshing assembly, traveling device and control levers.

The reel assembly with its rotary motion guides the rice plant to the cutting assembly. It consisted of five anchor bars, with each bar having 16 pieces of protruding prongs.

The machine's cutting device consisted of reciprocating cutting knives (**Figure 1**). It had a cutting width of 2000 mm. The height of cut was adjustable.

The rice combine's delivery device to its threshing unit was screw and a chain conveyor. The screw conveyor conveyed the cut plants to a chain conveyor which in turn delivered the cut plants to the threshing chamber of the combine.

The threshing assembly was of the axial-flow type and had an open-type threshing cylinder. The threshing cylinder had a length of 2010 mm and a diameter of 620 mm.

The cleaning assembly had a centrifugal fan which separated the chaffs, straws, empty grains and other light impurities from the threshed grain through air seams and an oscillating screen.

The rice combine was provided with grain tank (**Figure 2a**) which held approximately 500 L of grains. It had two outlet chutes (**Figure 2b**) where threshed grains were discharged for sacking.

The machine had a crawler-type traveling device made of rubber material.

The rice combine harvester was provided with control levers (**Figure 3**) at the left and center handles accessible to the operator. It had three classes of speed-change: low, medium and high both for forward and reverse.

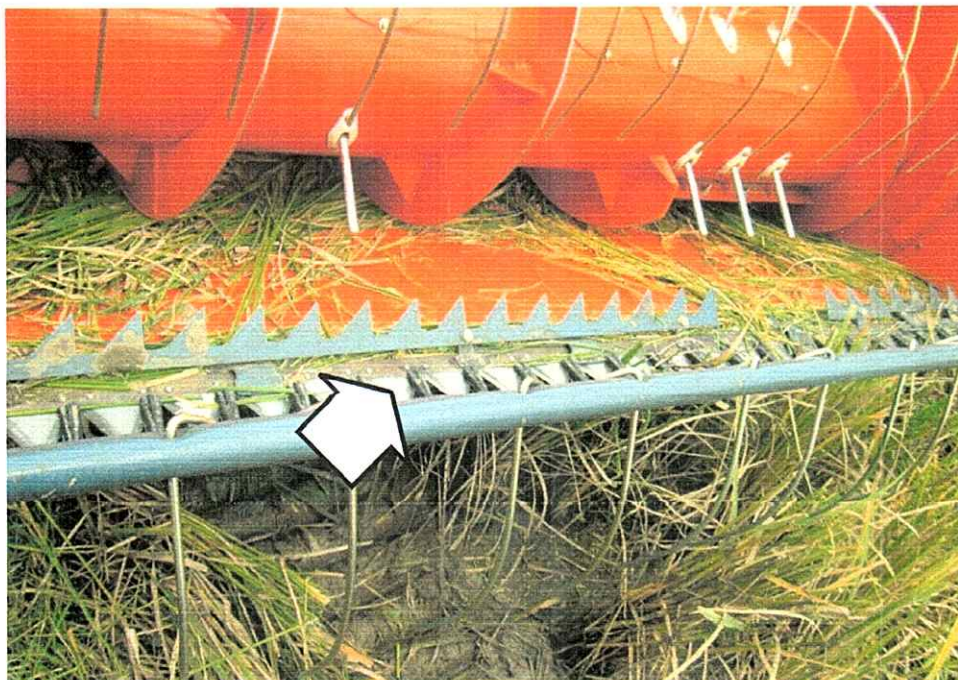


Figure 1. Reciprocating cutting knives



Figure 2. a) Grain tank
b) Outlet chutes

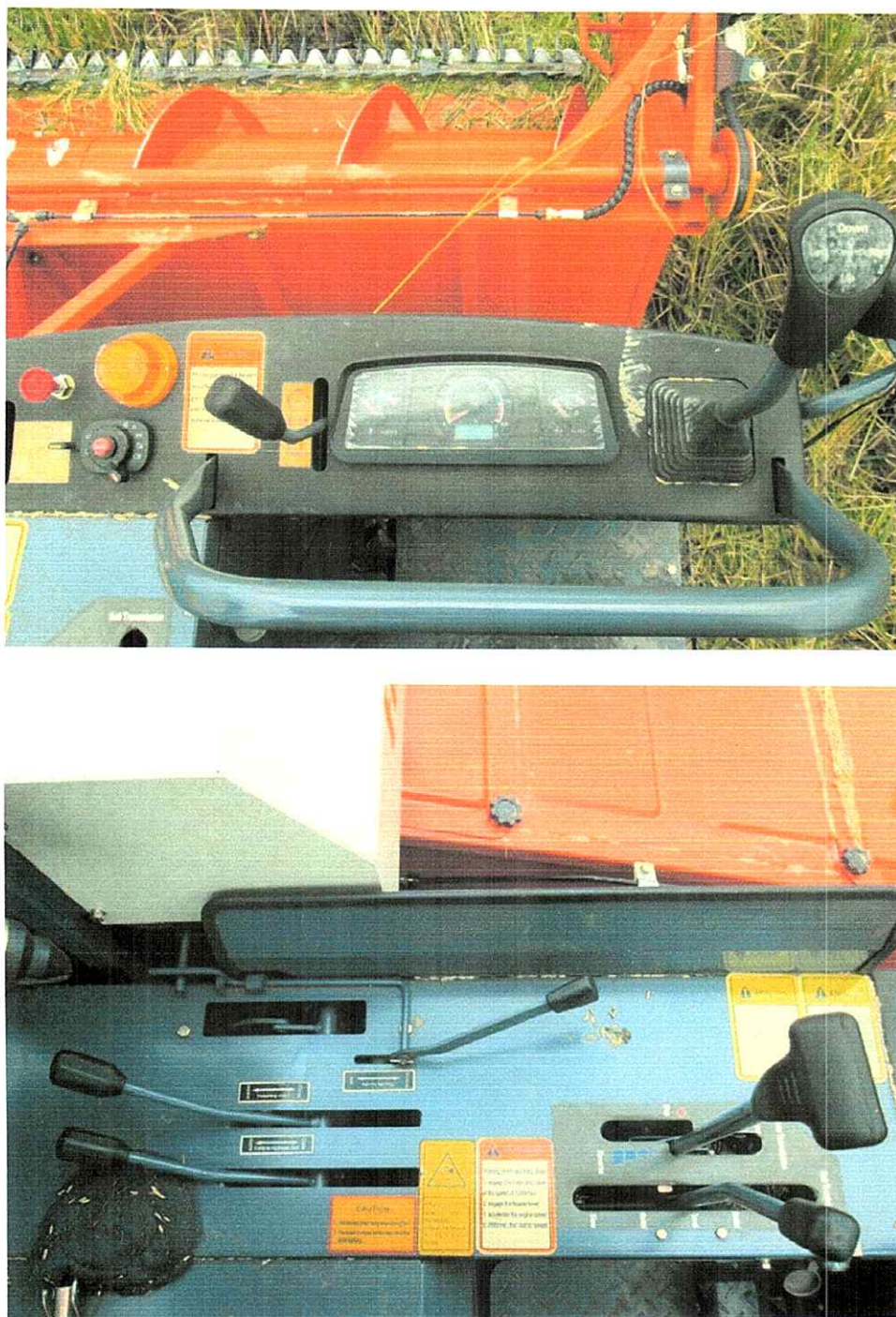


Figure 3. Control levers

MACHINE SPECIFICATIONS

Table 2. Manufacturer specifications of the FIELDKING FKMCCRHS-2.0 Rice Combine Harvester

1.	Brand	:	FIELDKING
2.	Model	:	FKMCCRHS-2.0
3.	Serial No.	:	ZZLGE216193
4.	Manufacturer	:	BERI UDYOG PVT., LTD. Plot no. 100-101, Sector 3, HSHDC, Karnal (Haryana), India
5.	Dimensions and weight		
5.1	Overall length (mm)	:	5150
5.2	Overall width (mm)	:	2890
5.3	Overall height (mm)	:	2700
5.4	Weight (kg)	:	3000
6.	Reel Assembly		
6.1	Reel dimensions, D × L (mm):		900 × 2028
6.2	No. of anchor bars	:	5
6.3	No. of prongs per anchor bar	:	16
7.	Cutting width (mm)	:	2000
8.	Threshing Assembly		
8.1	Type	:	Axial-flow
8.2	Dimensions, D × L (mm)	:	620 × 2010
9.	Cleaning System	:	3 centrifugal fans and one oscillating screen
10.	Rice Tank		
10.1	Capacity (L)	:	500
10.2	Outlet system	:	2 (Bagging)
11.	Traveling Device		
11.1	Type	:	Crawler
11.2	Ground contact, W × L (mm):		500 x 1700

Table 2. Manufacturer specifications of the FIELDKING FKMCCRHS-2.0 Rice Combine Harvester

12.	Maximum Traveling Speed (kph)		
12.1	Low speed	:	0 – 0.97
12.2	Medium Speed	:	0 – 1.85
12.2	High speed	:	0 – 2.50
13.	Rated Field Capacity (ha/h)	:	0.405 – 0.708
14.	Engine		
14.1	Brand	:	Not specified
14.2	Model	:	Not specified
14.3	Make	:	BERI UDYOG PVT., LTD. Plot no. 100-101, Sector 3, HSHDC, Karnal (Haryana), India
14.4	Type	:	4-stroke, water-cooled, vertical 4-cylinder, turbocharged diesel engine
14.5	Rated power (kW)	:	65.6 @ 2600 rpm
14.6	Displacement (cc)	:	2980

RESULTS AND DISCUSSIONS

Results of field performance tests are shown in **Table 3**.

Performance

The performance of the rice combine was evaluated in terms of its field capacity, field efficiency, output capacity, grain losses and grain quality.

Actual Field Capacity. It is the ratio of the area harvested and the total time of operation. The Field Capacity of the machine was 0.619 ha/h or 1.62 h/ha.

Field Efficiency. It is the ratio of the Actual Field Capacity and the Theoretical Field Capacity, expressed in percent. The Field Efficiency of the machine was 65.2%.

Header Loss. This refers to the grains that have fallen to the ground due to machine's cutting operation. Header Loss was 0.06%.

Separation/Blower Loss. This refers to the grains that come out of the threshing chamber together with the straw. Separation/Blower Loss was 0.10 %.

Unthreshed Loss. This refers to grains that remained on the panicles of the plant fed into the threshing chamber. There was no Unthreshed Loss.

Uncut Loss. This refers to the grains from the panicles that were not cut or missed by the reaping mechanism of the machine. There was no Uncut Loss.

Grain quality. Quality of shelled grain was determined in terms of percent cracked and broken grains. Results showed that the shelled output had a net percentage Cracked Grains of 1.33% and Broken Grains 0.06%. (Cracked grains refer to those grains whose endosperms are partially fractured while broken grains refer to those grains that are split into 2 or more pieces.)

Fuel Consumption

The 65.6 kW diesel engine had a Fuel Consumption Rate of 7.32 L/h or 12.7 L/ha.

Labor Requirement

During testing, at least three persons operated the rice combine (**Figure 4**), one for operating the machine and two for bagging the threshed grain.

Observations

1. The rice combine was easy to maneuver at corners.
2. The stability of the rice combine was maintained during operation.
3. The sacks of threshed grains were pushed to the ground after bagging.



Figure 4. Operators of the rice combine

FIELD PERFORMANCE TEST

Machine tested : FIELDKING Rice Combine Harvester
Model: FKMCCRHS-2.0

Location of test : Brgy. Maligaya, Muñoz, Nueva Ecija

Date of test : October 11, 2017

Table 3. Results of field performance test

ITEMS		TEST DATA
A. Condition of Crop		
1.	Variety	: “RC 222”
2.	No. of days after sowing	: 119
3.	Row Spacing	: Uneven
4.	Average Plant Height (mm)	: 1175
5.	Lodging angle of plant, degrees	: 90
6.	Plant population per square meter	: 20
7.	Grain-Straw Ratio	: 0.64
8.	Moisture Content of Grains (%)	: 24.3
B. Field Conditions		
1.	Area (m ²)	: 1500
2.	Type of Soil	: Sandy Loam
3.	Soil Resistance (kg/cm ²)	: 1.71
C. Field Performance		
1.	Method of Operation	: Circuitous
2.	Travelling Speed (kph)	: 4.75
3.	Effective Cutting Width (mm)	: 1200
4.	Height of Cut or Stubbles (mm)	: 392
5.	Actual Field Capacity	
a)	ha/h	: 0.619
b)	h/ha	: 1.62

Table 3. Results of field performance test (continuation)

ITEMS		TEST DATA
6.	Theoretical Field Capacity	
a)	ha/h	: 0.950
b)	h/ha	: 1.05
7.	Field efficiency (%)	: 65.2
8.	Threshed Grain Output Capacity (kg/h)	: 3006
9.	Purity (%)	: 93.2
10.	Header Loss (%)	: 0.06
11.	Separation/Blower Loss (%)	: 0.10
12.	Unthreshed Loss (%)	: Negligible
13.	Scattering Loss (%)	: Negligible
14.	Total Grain Loss (%)	: 0.16
15.	Noise level, db(A)	: 103.1
16.	Fuel consumption	
a)	L/h	: 7.32
b)	L/ha	: 12.7
D.	Laboratory Grain Analysis	
1.	Before Harvest	
a)	Cracked (%)	: 0
b)	Broken (%)	: 0
2.	After Harvest	
a)	Cracked (%)	: 1.33
b)	Broken (%)	: 0.06
3.	Net Percent	
a)	Cracked	: 1.33
b)	Broken	: 0.06

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